

Junyi (Conny) Zhou

Boston, MA 02215

github.com/JunyiZhou-Conny

Tel: (404) 663-5601

Email: junyizhou@hsph.harvard.edu

linkedin.com/in/junyi-zhou-270208247

EDUCATION

- **Harvard T.H. Chan School of Public Health** Boston, MA
M.S. in Health Data Science *August 2025 – December 2026*
 - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 *August 2021 – May 2025*
 - Honors: Dean's List Spring 2022, Spring 2023
 - Coursework: Machine Learning, NLP, Probability and Statistics, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, C++, Java, JavaScript
- **Health / ML Stack:** PyTorch, TensorFlow, Hugging Face, scikit-learn, scanpy, AlphaFold tooling
- **LLM in Clinical Settings:** RAG, agentic pipelines, prompt/context management, HIPAA-aware application design
- **Data & Cloud:** MongoDB, PostgreSQL, MySQL, AWS, Spark, HPC GPU workflows
- **Certifications:** AWS AI Practitioner; AWS Machine Learning Associate; AWS Cloud Practitioner

SELECTED EXPERIENCE

- **Airway Management Simulation Chatbot** Atlanta, GA
Full-Stack Engineer & Scrum Master — Emory Pediatric Hospital *January 2024 – Present*
 - Architected a clinical **agentic LLM + RAG** system with OpenAI models to support airway-management training while controlling hallucinations and protecting sensitive simulation data.
 - Built a **HIPAA-conscious** full-stack platform (Python/Flask, React, MongoDB, AWS) with real-time analytics of resident performance.
 - Used prompt engineering and context-window management for rapid-cycle deliberate practice scenarios.
 - Led a team of 6 with clinician feedback loops. Demo: CloudFront.
- **Cross-Species Drug Translation with Generative AI** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute *February 2026 – Present*
 - Developing Optimal Transport, VAE, and diffusion-based frameworks to model animal-to-human translation using scRNA-seq atlases.
 - Implementing custom **PyTorch** CellOT methods and scanpy pipelines for cell-type labeling and subpopulation matching (e.g., T-cell subsets).
- **AlphaFold Protein–Nucleic Acid Interaction Prediction** Atlanta, GA
Data Analyst — Bou-Nader Lab, Emory Department of Biochemistry *February 2025 – August 2025*
 - Built pipelines integrating AlphaFold-Multimer / AlphaFold3 for protein–nucleic acid complex prediction; automated GPU inference on HPC.
 - Filtered structures with pLDDT/PAE metrics for wet-lab follow-up; wrote CLI tools for non-computational researchers.
 - GitHub.
- **Transformer Architecture Reimplementation** Atlanta, GA
Machine Learning Researcher — Emory Department of Computer Science *October 2024 – January 2025*
 - Implemented Transformer components in PyTorch from scratch; achieved 9.38 BLEU on EN→DE translation after tuning.
 - GitHub.